* Statistics

1. What is statistics?
Statistics is the process of collecting, organizing, summarizing, analyzing and interpreting data to make conclusion.

2. Why do we study statistics?
- Summarize large amounts of data
- find patterns and relationships
- compare groups
- make evidence-based decisions
- support prediction and machine learning

3. main branch of Statistics.
- Descriptive statistics: describe the data we have, using tables, graphs, mean, median and other summaries.

- Inferential statistics: uses a sample to make estimates or conclusions about a large population.

4. Statistics in data science / machine learning.
Statistics helps summerize data, uncertainty, find relationship detect usual value, and evaluate model.

eg:-
A model can compare study with marks to explore whether more study time is associated with higher marks.

5. Data and frequency.

Data means collect facts or value. eg:-
marks 50, 60, 70, 80

Frequency means how many times a value occurs. In 50, 60, 60, 70, 80, the frequency 60 is 2.

A frequency table lists each value or class interval and its frequency, cumulative frequency is the running total of frequencies.

6. mean (average)

The mean is the sum of all values divided by the number of values.

mean = sum of value ÷ number of values

example : data = 60, 70, 80, 90, 100

mean = (60+70+80+90+100) ÷ 5 = 400 ÷ 5 = 80

mean is useful for summarizing numerical data, but unusually high or low value (outliers) can pull the mean toward them.

7) median

The median is the middle value after sorting values from smallest to largest.

odd number of values: 4, 7, 9 → median = 7.

Even number of values: 2, 4, 8, 10 → median = (4+8) ÷ 2 = 6.

For a frequency table, use cumulative frequency to identify the middle position (s).

8) mode

The mode is ~~value~~ the value that occurs most often. Example 2, 3, 3, 7 → mode 3.

A dataset can have one mode multiple modes, or no repeated mode.

9) Comparing mean, median and mode.

mean uses every value but is sensitive to ~~outline~~ outliers.

median is resistant to extreme value and is useful for skewed data such as salaries.

mode identifies the most common value and can be used for categories.

10) cumulative frequency

cumulative frequency (cf) is the running total of frequency.

11. Range

Range : Maximum value - Minimum value

eg:-

45, 50, 60, 80 → range = 80 - 45 = 35.

range is easy to calculate, but it only uses the smallest and largest values.

* variance.

variance describe how far value spread around the mean. A small variance means values are close to the mean a large variance means value are more spread out.

population variance

$$\sigma^2 = \Sigma(x-\mu)^2 \div N$$

sample variance

$$s^2 = \Sigma (x-\bar{x})^2 \div (n-1)$$

symbols: x = each value; μ population mean; $\bar{x}$ = sample mean; N = population size; n = sample size; Σ means add all items.

* standard deviation (SD)

standard deviation is the ~~process~~ square root of variance. It describe the typical spread of data around the mean and uses the same units as the Dryinal data.

population SD: $\sigma = \sqrt{\sigma^2}$ | sample S.D.'s $= \sqrt{s^2}$

* Interquartile range (IQR)

$$IQR = Q3 - Q1$$

Q1 is the first quartile (25th percentile)
Q2 is the median (50th percentile). Q3 is the third quartile (75th percentile)
IQR measure the spread of the middle 50% of data.

* percentiles

A percentile indicates the position below which a given percentage of observation falls. The 25 ~~per~~ percentile is Q1, the 50th percentile is the median and the 75th percentile is Q3.

For an ordered dataset, one common ~~intro~~ introductory position formula is: position = $p/100 \times (n+1)$ where p is the desired percentile. Different textbooks software may use different percentile convention, so follow the method.

* quartiles.

Q1: First quartile approximately 25% of observation are at below it.

Q2: second quartile the the median approximately 50% are at below it.

Q3: third quartile approximately 75% are at or below it.

* Detecting outliers using IQR

lower fence : $Q1 - 1.5 \times IQR$

upper fence : $Q3 - 1.5 \times IQR$

values below the lower fence or above the upper fence are potential outliers. An outliers is not automatically an error; check whether it is a real observation or a data entry problem.

* Box Plot

Box plot shows the minimum / low end Q1 median Q3 and maximum / high end often including outliers shown as seprate points).

* symmetric distribution.

A symmetric distribution has a similar shape on both side of its centre. In a perfectly symmetric, unimodel distribution mean and median are often equal or close.

* skewed distribution.

Right-skewed (positive skew): the long tail extend to the right. A few high values may pull the mean above the median.

* Covariance :

Covariance describes whether two numerical variable tend to move together. positive covariance means they trend to increase together; negative covariance mean one trends to increase as the decrease; covariance near zero suggests little linear co-movement.

Sample covariance

$$= \Sigma[(x - \bar{x})(y - \bar{y})] \div (n-1)$$

* what is correlation

correlation mesure is the strength and direction of a linear relationship between two ~~number~~ numerical variable. The correlation coefficient r is between -1 and +1

- $r = +1$: perfect positive linear relationship
- $r = -1$: perfect negative linear relationship
- $r \approx 0$: no or weak linear relationship (a non-linear may still exist).

* Types of correlation.

- strong positive: points lie close to an upward -sloping line.
- weak ~~negative~~ positive : upward trend with more scatter.
- strong negative! points lie close to a downward -sloping lie.
- weak negative! slight downward trend, zero linear correlation: no clear straight pattern.
- correlation is based on covariance and

Standard deviation.

$r = cov(x,y) \div (sx \times sy)$

- correlation has no units, unlike covariance, always inspect a scatter plot and watch for outliers.

* Z-score

A z-score tells how many standart deviation a value is above or below the mean.

$z = (x - \mu) \div \sigma$

x = observed value, μ = mean, σ = standard deviation.

* Standardization vs normalization.

Standardization transform values using the mean and standard deviation, often producing z-score with mean near 0 and SD near 1.

z-score standardization : $z = (x - mean) \div SD$

min-max normalization rescales values to a chosen range, often 0 to 1.

min : $x' = (x - minimum) \div (maximum - minimum)$

* Normal distribution.

The normal distribution is a symmetric bell-shaped probability distribution its means, median and mode are equal at the centre.

* Normal distribution formula (concept)

A normal distribution is described by its mean μ and standard deviation σ. The formula gives the probability density at a value; for beginner use, understand the bell shape, centre, and spread before memorizing the formula.

* Normal distribution and z-scores

A z-score converts a value into standard deviation units - The helps compare score measured on different scales and find approximate probabilities using a standard normal table or software.

* Outliers and normality.

A value far from the mean standard-deviation units may be unusual. For approximately normal data, value more than 3 SD from the mean are uncommon, but outliers description should also consider context and data quality.

* Statistical inference and hypothesis testing.

Statistical uses sample data to estimate population characteristic or test a claim. many common method assume normality residule or sampling distribution, depending on the method.

- t-test : compare means, commonly for one or two groups.
- Regression : models relationships between variable common inference relies on asumption about respduals.
- confidence interval : give a range of plausible value for a ~~probability of~~ parameter under ~~result at least as~~ assumption.
- p-value : under the null hypothesis and model assumption tha probability of obtaining rasut at least extreme observed

* where normal distribution is useful. It appears in many measurement and statistical - modeling setting and it used in quality control, test score confidence interval and some machine - learning method. Real-world data are not always normaly distributed.

* What is probability.
probability measure how likely an event is range from 0 to 1 or 0% to 100%.
- 0 = impossible
- 0.5 = 50% chance
- 1 = certain

* Basic probability.
P(A) = Number of favourable outcome / total equally likely outcomes.

* probability with playing cards.
A standard deck has 52 cards, 4 suits with 13 cards each. Assuming a well shuffled check:
P(drawing Ace) = 4/52 = 1/13 ≈ 7.69%.
P(not drawing an Ace) = 48/52 = 12/13 ≈ 92.3%

* complement rule
$$P(\text{not } A) = 1 - P(A)$$

* Addition rule
for mutually exclusive events (they cannot happen at the same time):
$$P(A \text{ or } B) = P(A) + P(B)$$

* Multiplication rule and independent rule.
for independent event the occurence or an desnot change the probability of the other.
$$P(A \text{ and } B) = P(A) \times P(B) \text{ (if independent)}$$

* conditional probability

conditional probability is the probability of event A given the event B has occured.

$P(A|B) = P(A \cap B) \div P(B)$, provide $P(B) > 0$

* Bayes' theorem

Bayes theorem update the probability of a cause hypothesis after observing evidence

$P(A'|B = [P(B|A) \times P(A)] \div P(B)$

* probability quick rules.

- $0 \leq P(A) \leq 1$.
- $P(S) = 1$ for the full sample space.
- $P(\phi) = 0$ for an impossible event.
- $P(\text{not } A) = 1 - P(A)$
- mutually exclusive : $P(A \text{ or } B) = P(A) + P(B)$.
- independent : $P(A \text{ and } B) = P(A) P(B)$.
- conditional : $P(A|B) = P(A \cap B) / P(B)$.
- Bayes : $P(A|B) = P(B|A) P(A) / P(B)$.